

Om Patel

Atlanta, GA | (774) 368-9850 | opatel97@gatech.edu | US Citizen | linkedin.com/in/ompatel25

Education

Georgia Institute of Technology

B.S. in Electrical Engineering

May 2028

GPA: 4.00

- Concentration: Circuit Technology and Robotics and Automation

Experience

Comark Corporation

Electrical Engineering Co-op: Research & Development Team

Milford, MA

May '24 - May '25

- Programmed **Embedded C firmware** for 3 automated testing systems that evaluated **2000+ product types**, improving defect detection by **20%** & reducing overall QA time by **+40%**.
- Designed **5 custom PCBs** using **KiCAD** & produced 35 custom cables using **SolidWorks** drawings, working cross-functionally to integrate new features & fixes across 150 product units & supporting 4 major client projects.
- Performed **oscilloscope, multimeter, thermal chamber**, & electromagnetic interference (MIL-STD-810, MIL-STD-461) testing, accelerating approval of 15 submarine-grade displays by **23%**.

Projects

GridGuard

Hardware Lead, Energy Hack @ Georgia Tech

Atlanta, GA

Jan. '25 - Present

- Designed custom power distribution circuits achieving **+90% efficiency**, utilizing **buck converters** to step down 12V to 5V for logic components & stabilize the battery reserve system during simulated outages.
- Led **3 rapid hardware prototyping iterations** over 48 hours, validating telemetry concepts on **ESP32** modules before migrating to an **Arduino** architecture to ensure **100% data reliability** during the final demo.
- Integrated voltage & current sensors with a Next.js dashboard powered by **AI weather prediction**, enabling the system to autonomously trigger load-shedding switches with **<200ms latency** during high-risk events.

Touch Screen Tester

Project Lead at Comark Corporation

Milford, MA

May '24 - Oct. '25

- Wrote **Python** scripts & **G-code** to engineer an automated electro-mechanical test system that evaluated device reliability & reduced QA time by **94%** (8 hours to 30 mins).
- Designed a web-based interface with **customizable test parameters & data queries** to output characterization studies (CSV), enabling engineers to diagnose anomalies (missed touches, double clicks) across **600+ products**.

Leadership/Activities

The Hive Makerspace

Prototyping Instructor (PI)

Atlanta, GA

Oct. '25 - Present

- Mentored 30+ peers weekly on building & commissioning prototypes, debugging & diagnosing issues with complex **electro-mechanical implementations** involving **sensors**, feedback loops, & **motor control**.
- Instructed a **~450-student cohort** in low-level hardware programming & logic design, diagnosing complex **VHDL implementations**, resolving **Assembly/C firmware bugs**, & validating theoretical circuit analysis on physical hardware.
- Guided users through full-stack prototyping workflows, debugging **custom PCB designs & validating firmware** logic in **C & Python** to ensure successful integration of analog & digital subsystems.

Energy Club

Technical Project Committee

Atlanta, GA

Aug. '25 - Present

- Engineered **AC-to-DC rectification circuits & micro-capacitor storage units**, capturing intermittent piezoelectric pulses from **30+ energy-harvesting tiles** to develop scalable sustainable power prototypes for campus-wide deployment.
- Designed custom **PCBs & integrated microcontrollers** to convert raw sensor output into usable energy, improving system efficiency by **25%** compared to baseline models.

Skills

Programming: Python (Scripting & Automation), C, C++, MATLAB, JavaScript, Node.js, HTML, Verilog/VHDL

Design: Electro-mechanical Systems, Analog/Digital Circuits, PCB Design/Debug, Microcontrollers, Sensors/Feedback

Software: LTspice, Cadence OrCAD & Allegro, SolidWorks, Multisim, Autodesk Inventor (Certified), AutoCAD, KiCAD

Hardware & Testing: Reliability Testing Setup, Soldering & Rework, Oscilloscope, Logic Analyzer, Multimeters, FPGAs

Workplace Efficiency: Cross-disciplinary Collaboration, Agile Project Management, OSHA 10-Hour, Six Sigma White Belt

Communication: Engineering Notebook, Technical Reports, Operating Procedures, Dimensioned Drawings

Honors: 4th @ GT Energy Hack, SkillsUSA National Top 10 (x2) – Mobile Robotics, BVT Varsity Math Team Champion (x2)